RAW AND QCOH PUSHFORWARDS AGREE AT FLAT OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- delivered companions: direct-image proposition, adjoint-kernel; chapter definition of parasitic. delivered companions proposition-direct-image-quasi-coherent, lemma-adjoint-kernel-parasitic; same-chapter definition-parasitic (definition, no proof needed). Whole argument.

Source reference. Original proposition 077A: sources/stacks-chow/tags.txt, source lines 9195–9195. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0773: sources/stacks-chow/tags.txt, source lines 9188–9188. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. Let $y : V \to \mathcal{Y}$ in $\mathrm{Ob}(\mathcal{Y})$ with y a flat morphism. Let $\mathcal{F}$ be in $QCoh(\mathcal{O}_\mathcal{X})$. Then $(f_*\mathcal{F})(y) = (f_{QCoh,*}\mathcal{F})(y)$ and $(R^i f_*\mathcal{F})(y) = (R^i f_{QCoh,*}\mathcal{F})(y)$ for all $i \in \mathbf{Z}$.*

Proof. This follows from the construction of the functors $R^i f_{QCoh,*}$ in Proposition 077A, the definition of parasitic modules in Definition 0773, and Lemma 0779 part (2). □

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